

Standardization of Materials R&D Data Schema and Keywords

Kwang-Ryeol Lee (krlee@kist.re.kr)

**Computational Science Research Center
Korea Institute of Science and Technology**

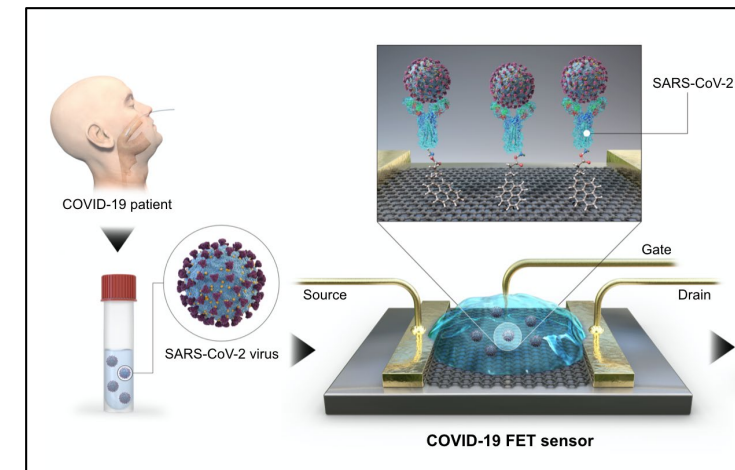
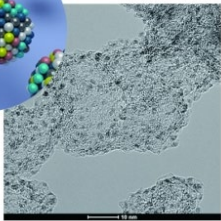
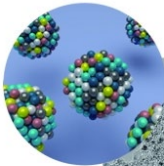
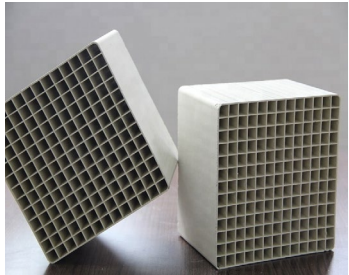
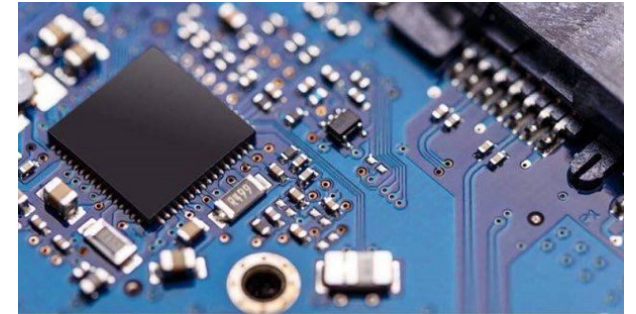
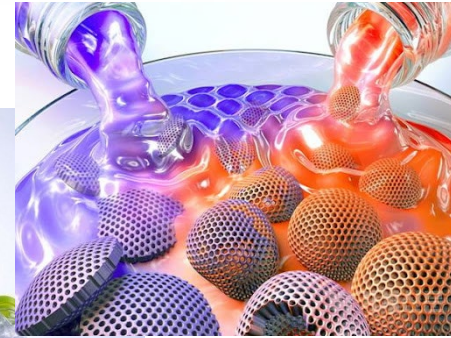
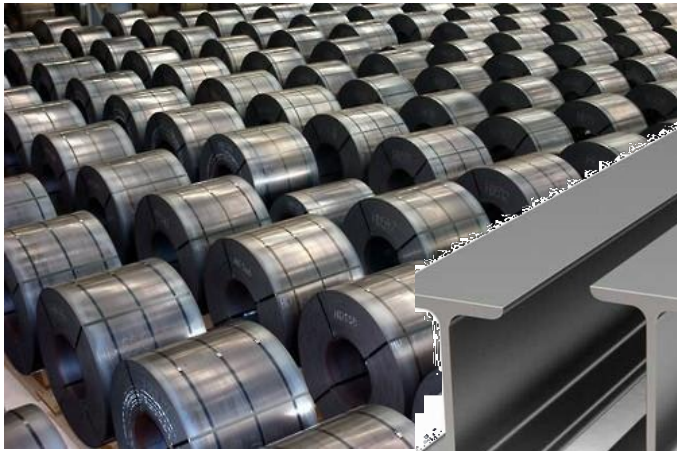
**Materials R&D Data Standard Expert Committee
National Center for Materials Research Data**

5 Key Points of the Standard

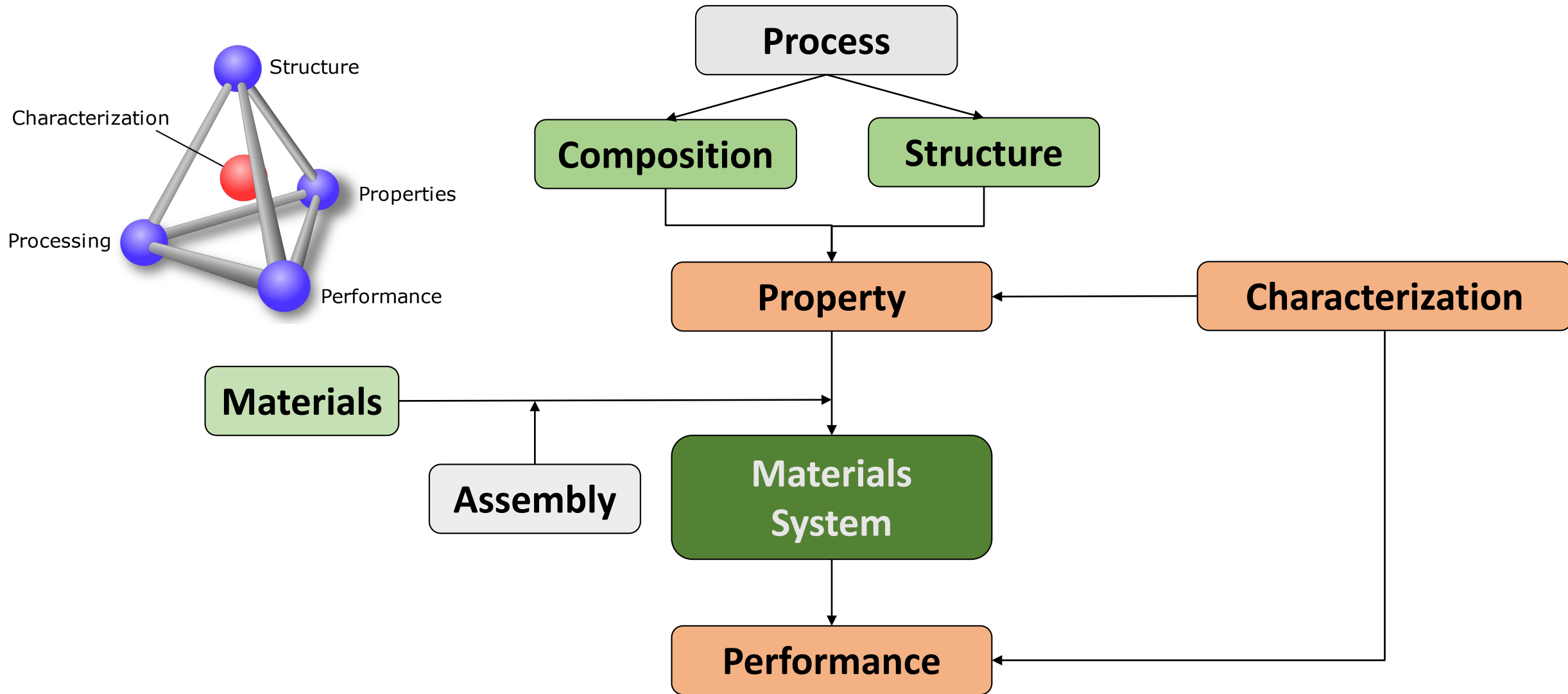
1. The committee has developed **a harmonized materials data schema** to ensure consistent management of materials research data across various sub-disciplines.
2. The committee aims to standardize data schemas and keywords that encompass **the Process-Structure-Property-Performance (PSP) relationship** across a wide range of materials fields.
3. This goal is achieved by **separating intrinsic materials data from application-specific data**, through the introduction of the **“materials system”** concept within the standard schema.
4. Our approach structures the data to reflect typical **experimental and theoretical materials research workflows**, facilitating easier adoption by researchers.
5. The committee periodically publishes an undated dictionary of keywords and schema definitions twice a year on their GitHub repository: **<https://github.com/NCMRD/Standard-Expert-Committee>**.

Common Data Structure for Wide Spectrum of Materials?

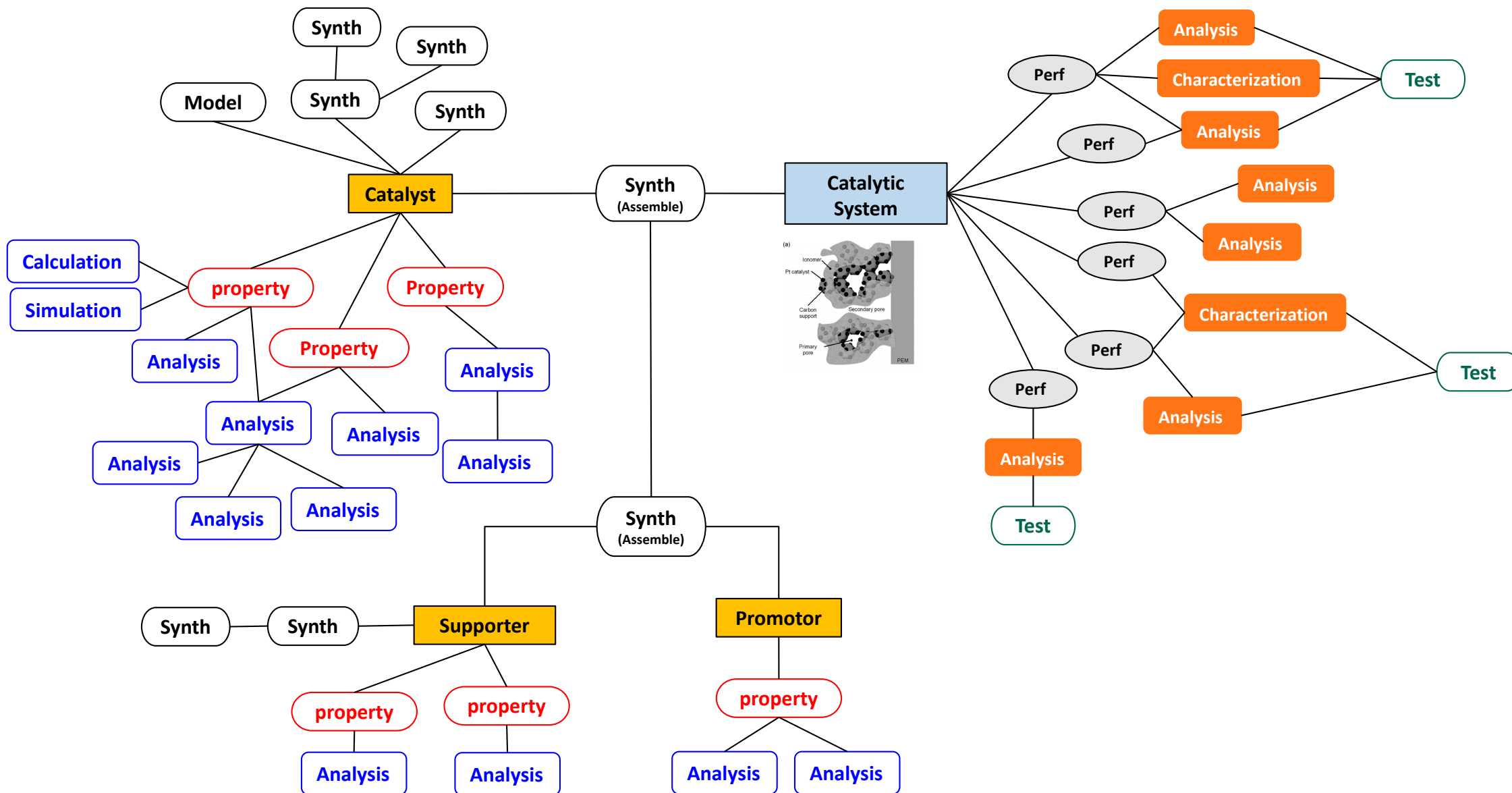
森羅萬象



Harmonized AI-ready Materials R&D Data



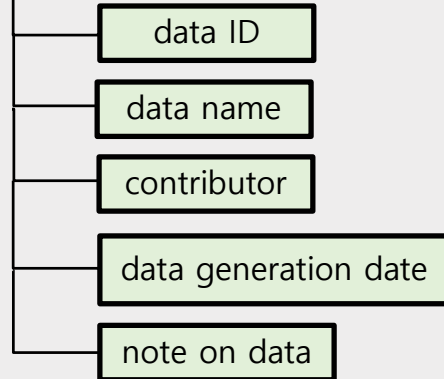
KIST
Korea Institute of
Science and Technology



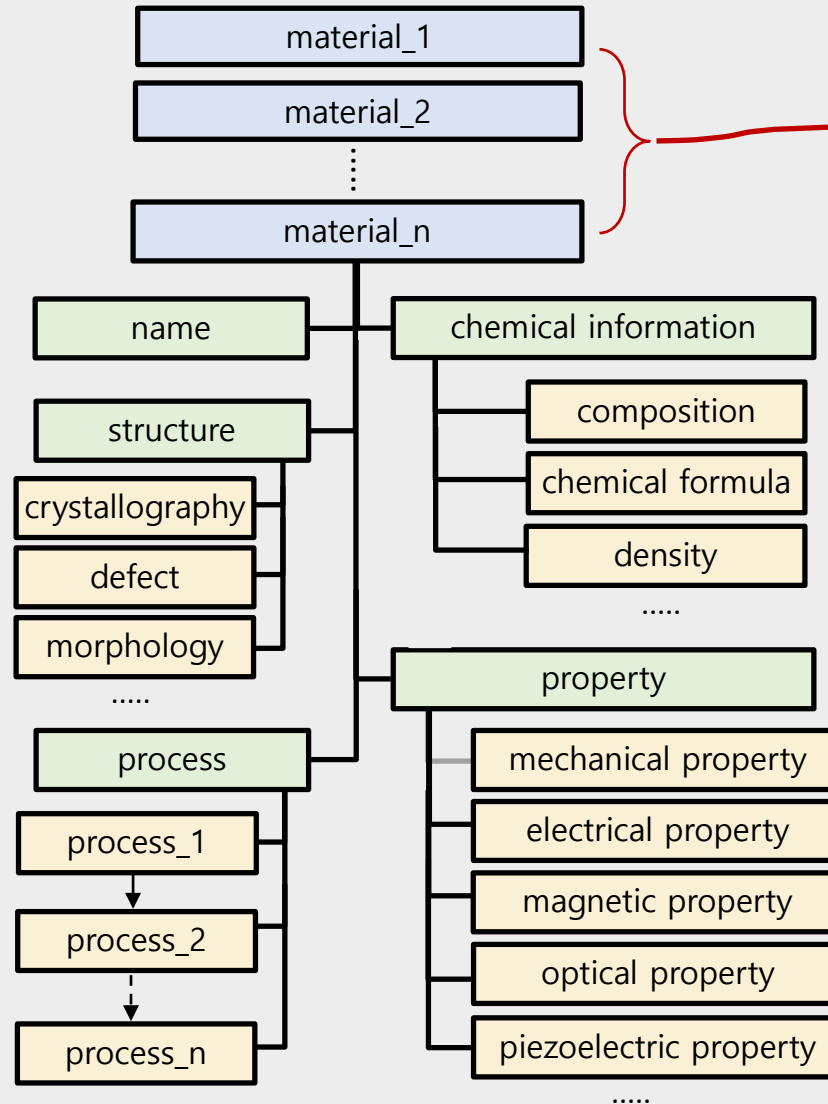
Materials Research Data Set

45T

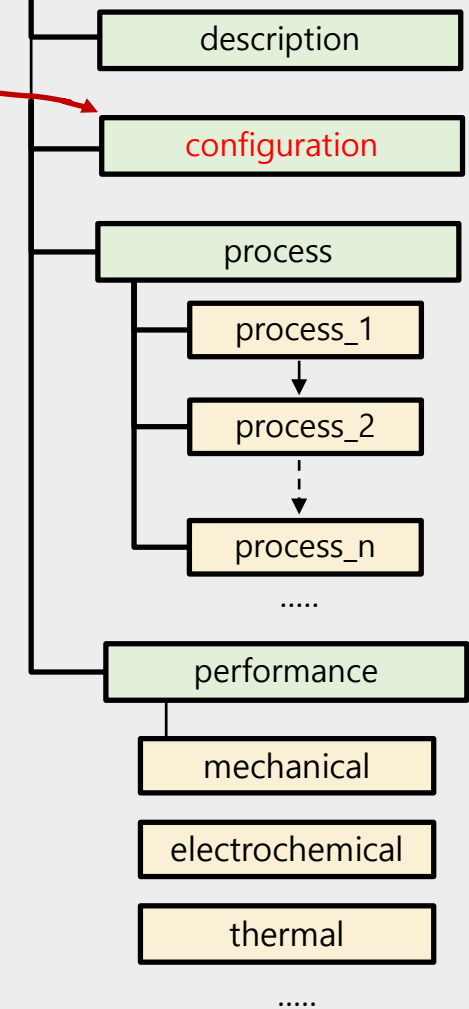
Metadata



Materials



System



Standard Structure of Materials Data Set

metadata

Meta data section of the data set

- data ID
- data name
- contributor
- data generation date
- note on data

materials_{*n*}

*Data section of each *n* materials constructing the materials system
Each materials' data is indexed by the system variable "Materials_{*n*}"*

- name
- chemical information
- process
- model
- property

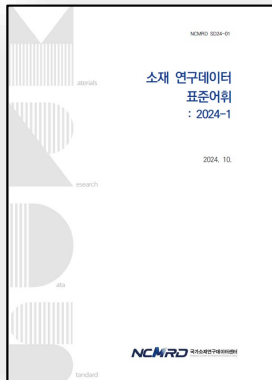
system

Data section of Materials System

- description
- configuration
- process
- performance

```
▼ 0 {4}
  ▼ meta {5}
    data ID : DOI:111.222.333.444
    data      : Pt3Ni particle on graphite support data
    name      : 1
    ► contributor {2}
    data generation date : 2020-04-28
    note on   : information of material_2 comes from
    data      : Aldrich catalog\n10.1021/n1401881z
  ▼ material_1 {4}
    name : Pt3Ni
    ► chemical information {2}
    ▼ process [3]
      ▼ 0 {1}
        ► solvothermal {8}
      ▼ 1 {1}
        ► heat treatment {10}
      ▼ 2 {1}
        ► centrifugation {5}
    ► property {1}
  ► material_2 {2}
  ▼ system {4}
    description : Pt3Ni/C catalyst system for CO2
    reduction
    ▼ configuration {3}
      active material : Material_1
      amount of active material : 0.003
      support material : Material_2
    ► process [3]
    ▼ performance {1}
      ▼ electrochemical {1}
        ▼ ORR {3}
          ▼ area-specific activity {3}
            value : 49
            uncertainty : 0.1
          ▼ measurement {1}
            ► electrochemical activity {5}
          ► mass-specific activity {3}
          ► Faradaic efficiency {3}
```

Outcome: Structured Data Keyword Dictionary



Product Details

Name:	Adjustable Race	List Price:	\$0.00
Product Number:	AR-5381	Standard Cost:	\$0.00
Model:		Sell Start Date:	1/06/1998
Category:		Sell End Date:	<d/MM/yyyy>
Subcategory:		Discont. Date:	<d/MM/yyyy>
Product Line:		Safety Stock:	1000
Class:		Reorder Point:	750
Style:		Days To Man.:	0
Color:		Make:	☐
Size:		Finished Goods:	☐
Weight:			

OK Cancel

K-MDS

About Data Search Support My Page

Material innovation research To create an ecosystem

Korea Materials Data Station

K-MDS is a materials research data collection, sharing, and utilization platform established and operated at the national level.

We want to contribute to creating a materials data-based research and innovation ecosystem in Korea.

Materials Property Composition

Please enter your search terms. (Use the special character)

Recommended keywords #Material #124912491

CATEGORY Data type

Catalyst Porous materials Memristive Photodetector Gas sensor Material

UTILIZATION Top search data Latest registration data Popular service

<https://github.com/NCMRD/Standard-Expert-Committee>

Korea Materials Data Station

Contents of the Structured Vocabulary Dictionary

- ▶ metadata vocabulary : { 8 props }
- ▶ materials vocabulary : { 1 prop }
- ▶ system specific vocabulary : { 14 props }
- ▶ common vocabulary for analysis method : { 67 props }
- ▶ common vocabulary for materials process : { 70 props }
- ▶ numeric data expression : { 2 props }

Welcome your participation for extending
the standard schema and keywords!

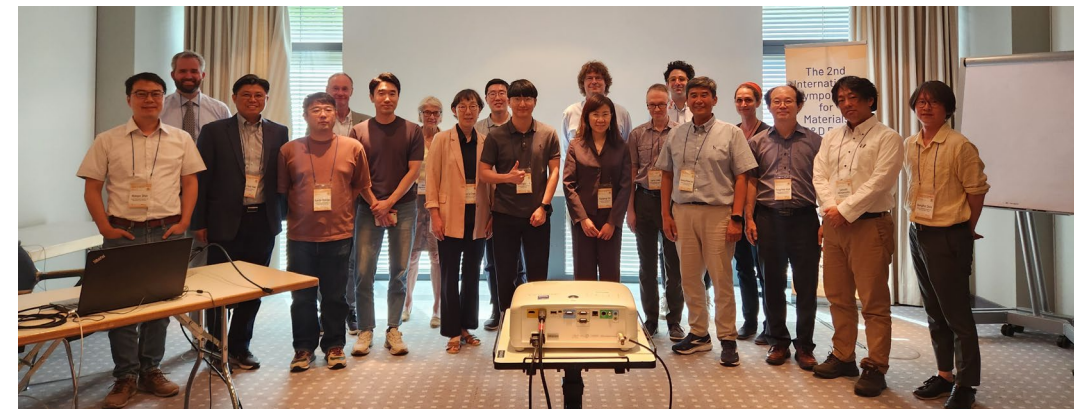
mailto: krlee@kist.re.kr

어휘군	category 1	category 2	category 3
metadata vocabulary	data name		
	contributor	name affiliation	
	data generation date		
	note on data		
materials common vocabulary	materials...(n)	name	
		chemical information	composition crystallography SMILES
		process	(process..n)
		model	elements
			state
			box dimension
			periodic boundary condition
			structure file
		property	structural property
			mechanical property
			electrical property
			magnetic property
			thermodynamic property
			thermal property
			optical property
			corrosion property
system specific vocabulary	system_catalyst	description	
		configuration	active material
			amount of active material
			promotor
			amount of promotor
		process	support materials
			(process..n)
	system_porous_materials	performance	electrochemical thermal
		description	
		configuration	framework...(n) inserted molecule
		process	(process..n)
	system_memristi	performance	gas adsorption
		description	

KIST
Korea Institute of
Science and Technology



- July 17~18, 2025
- NIMS Campus, Tsukuba, Japan
- Co-organized by NCMRD, Korea and NIMS, Japan
- <https://ncmrd-symposium.net>



2nd Symposium (2023, July, Wiesbaden, Germany)

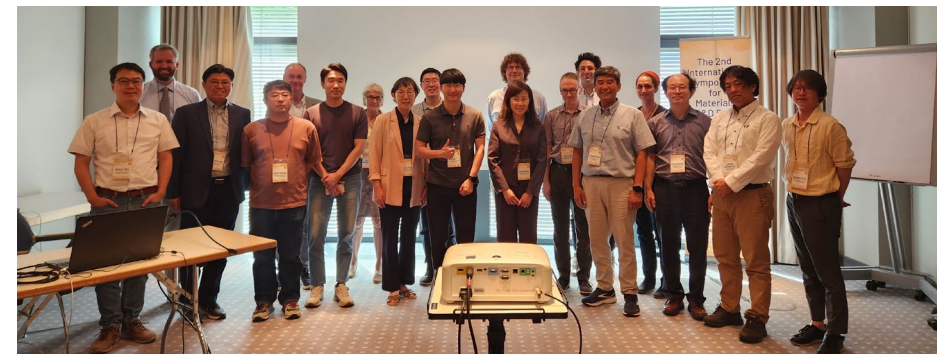


3rd Symposium (2024, Jeju, Korea)

Acknowledgement

- **National Center for Materials Research Data**
- **Contributors**

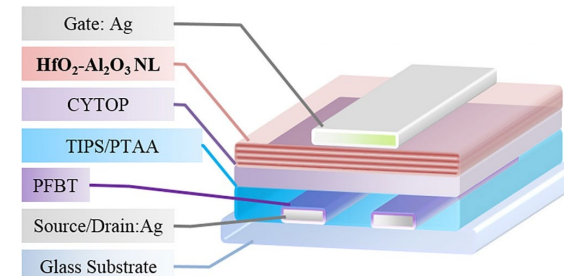
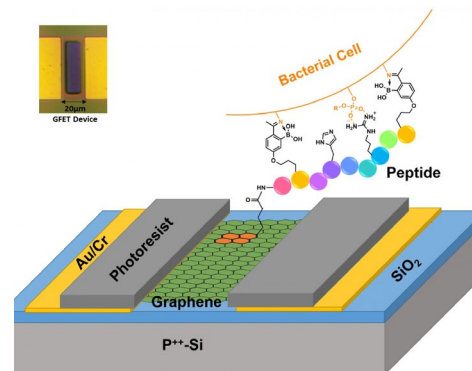
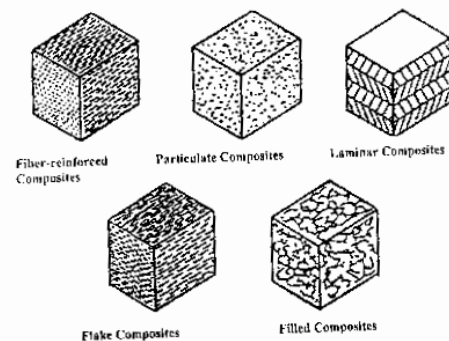
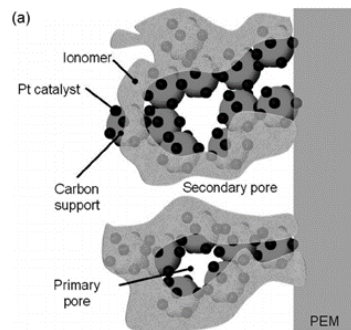
허용학 (KRISS), 장현주 (KRICT), 한승우 (서울대), 김용주 (국민대), 심형석 (KRISS), 한상수 (KIST), 신호선 (KRISS), 오창석 (KIMS), 이정훈 (KIST), 이병주 (KIST), 김지한 (KAIST), 김현유 (충남대), 서동화 (KAIST), 임경근 (KRISS), 이동화 (POSTECH), 박선화 (KRISS), 김수현 (KIMS), 장효선 (KIMS), 김정한 (한밭대), 김정환 (한밭대), 안재평 (KIST), 김낙균 (KIST), 박수형 (KIST), 제송근 (전남대), 명승택 (세종대), 권용우 (홍익대), 김범수 (KRICT), 류권상 (아이피스), Eva Unger (Humboldt U), 이에리 (KRICT), 정증현 (KIST), 최환진 (KBSI), 권기창 (KRISS), 방기훈 (KIST), 이정현 (고려대), 유재상 (KIST), 김재석 (KRISS), 이우 (KRISS), 이상한 (GIST), 정인 (서울대), 김일호 (철도대), 손재성 (POSTECH), 이규형 (연세대), 유태수 (충북대), 이상한 (GIST), 송재용 (POSTECH), 김선태 (전북대), 이관영 (고려대), 이병주 (POSTECH), 임영목 (KIMS), 차필령 (국민대), 최우진 (KRICT), 한흥남 (서울대), Xiaowei Li (CUMT), Aiyang Wang (NIMTE), Haiqing Yin (USTB), ...



Definition of Materials System

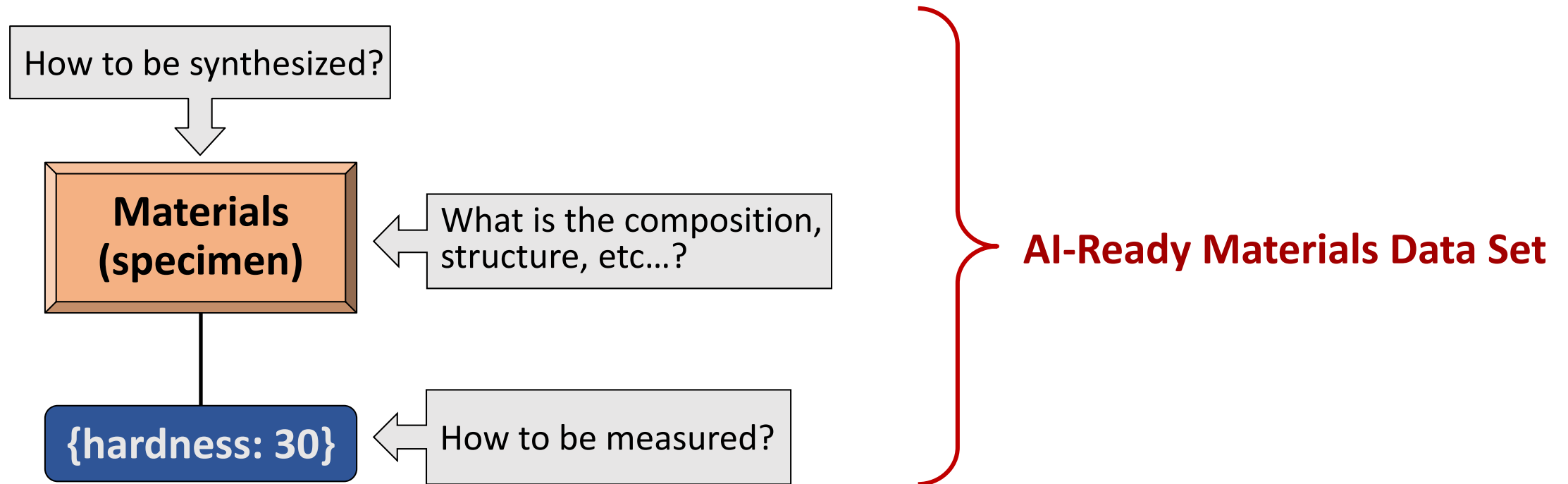
- Object that is composed of one or more materials to characterize the performance of a material.

Materials System Examples	Materials 1	Materials 2	Materials 3	Materials 4	...
Catalytic Materials	Catalyst	Supporter	Promotor		
Composite Materials	Fiber	Matrix			
Sensor Materials	Sensor	Electrode	Substrate	Protective Layer	
Cathode in Secondary Battery	Cathode 1	Cathode 2	Electrode	Electrolyte	
Chrome Coating Steel	Substrate Steel	Cr Coating Layer			
Special Steel	Special Steel				

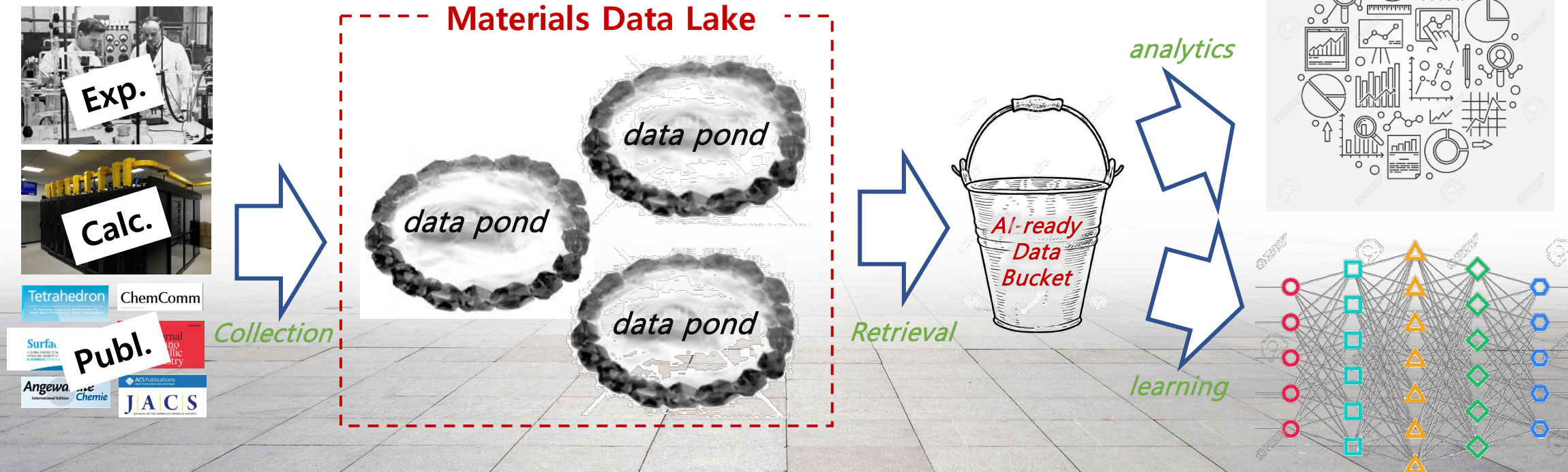


AI-ready Data for Materials Design

- *Explicitly Defined Data*
- *Understandable Context of the Data* in the viewpoint of materials science

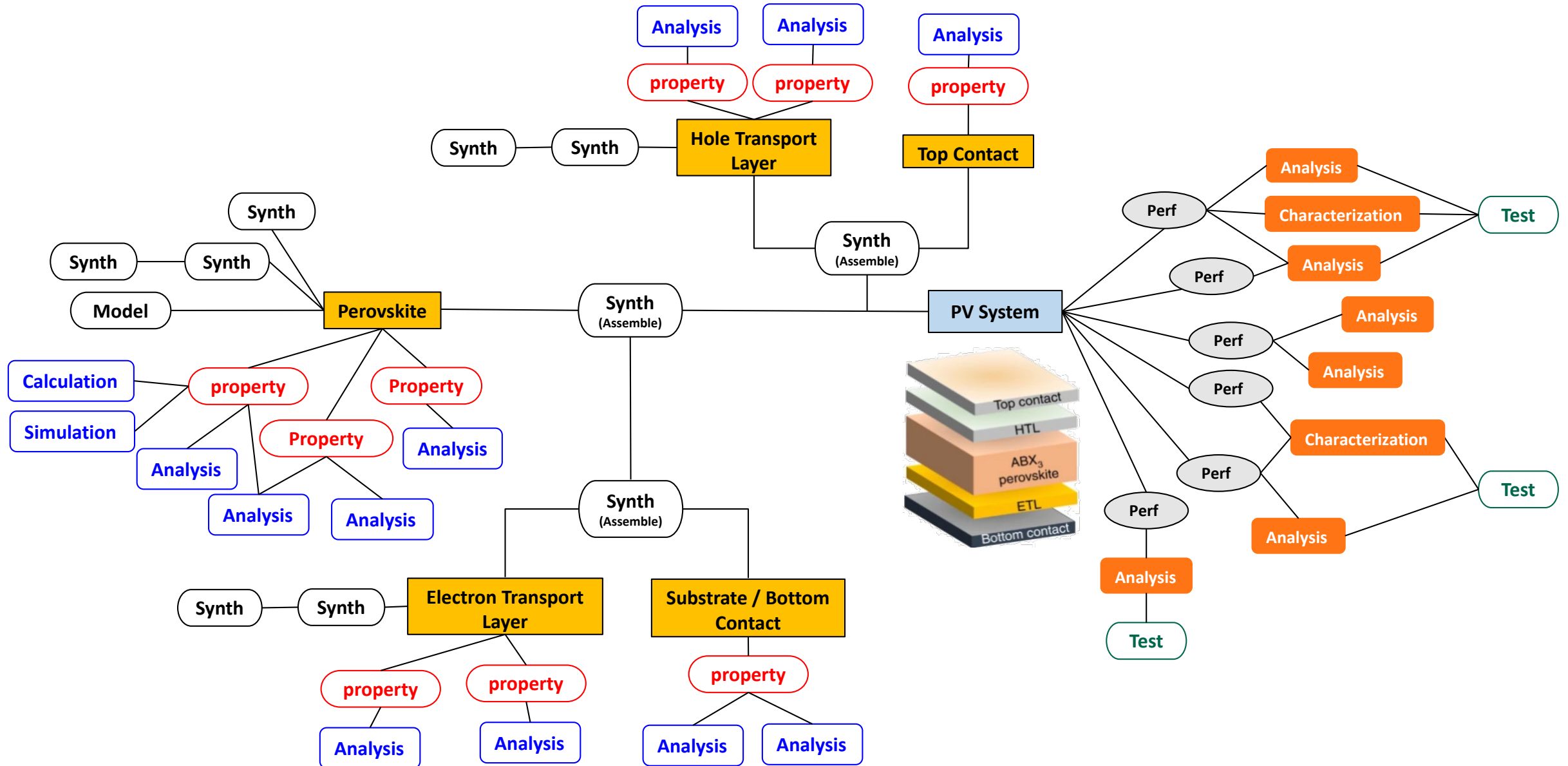


Rock Base for the Data Driven Materials Design



Standard Data Schema and Keywords
Language for Research with Materials Data

KIST
Korea Institute of
Science and Technology



Materials Vocabulary *(as of 2024.10, 12 groups of property)*

```
▼ materials vocabulary : {  
  ▼ material_(n) : {  
    eng_definition : materials that can be a component of materials system  
    alias : value  
    ▶ name : { 4 props }  
    ▼ chemical information : {  
      eng_definition : atomic scale information of materials  
      alias : Stirring time  
      ▶ composition : { 4 props }  
      ▶ purity : { 5 props }  
      ▶ density : { 5 props }  
      ▶ CAS number : { 4 props }  
      ▶ chemical formula : { 4 props }  
      ▶ structural formula : { 4 props }  
      ▶ SMILES : { 3 props }  
      ▶ molecular weight : { 4 props }  
      ▶ additive : { 4 props }  
    }  
    ▶ model : { 7 props }  
    ▶ structure : { 14 props }  
    ▼ property : {  
      eng_definition : attribute, quality, or chracteristic of materials  
      alias : value  
      ▶ corrosion property : { 18 props }  
      ▶ electrical property : { 34 props }  
      ▶ electrochemical property : { 3 props }  
      ▶ ferroelectric property : { 5 props }  
      ▶ gas transmission property : { 4 props }  
      ▶ magnetic property : { 36 props }  
      ▶ mechanical property : { 12 props }  
      ▶ optical property : { 10 props }  
      ▶ piezoelectric property : { 7 props }  
      ▶ rheological property : { 5 props }  
      ▶ thermophysical property : { 24 props }  
      ▶ tribological property : { 3 props }  
    }  
    ▶ process : { 3 props }  
  }  
}
```

```
▼ electrical property : {  
  eng_definition : physical properties that a material exhibits upon the application  
    of electric forces  
  alias : value  
  ▶ band gap bowing parameter : { 5 props }  
  ▶ band gap energy : { 5 props }  
  ▶ band gap type : { 3 props }  
  ▶ band gap graded : { 3 props }  
  ▶ carrier concentration : { 5 props }  
  ▶ carrier diffusion length : { 5 props }  
  ▶ carrier lifetime : { 5 props }  
  ▶ conduction band minimum : { 5 props }  
  ▶ charge of constituents : { 5 props }  
  ▶ critical temperature : { 5 props }  
  ▶ dielectric constant : { 5 props }  
  ▶ dielectric strength : { 5 props }  
  ▶ electric susceptibility : { 5 props }  
  ▶ electrical conductivity : { 5 props }  
  ▶ electronic density of states : { 6 props }  
  ▶ electron effective mass : { 5 props }  
  ▶ electron mobility : { 5 props }  
  ▶ exciton binding energy : { 5 props }  
  ▶ Fermi level : { 5 props }  
  ▶ Hall coefficient : { 5 props }  
  ▶ Hall voltage : { 5 props }  
  ▶ hole effective mass : { 5 props }  
  ▶ hole mobility : { 5 props }  
  ▶ ionic conductivity : { 5 props }  
  ▶ ionic mobility : { 5 props }  
  ▶ ionic activation energy : { 5 props }  
  ▶ permittivity : { 5 props }  
  ▶ resistivity : { 5 props }  
  ▶ valence band maximum : { 5 props }  
  ▶ spin-Hall angle : { 5 props }  
  ▶ spin-orbit splitting energy : { 5 props }  
  ▶ temperature coefficient of resistance : { 5 props }  
}
```

System Specific Vocabulary *(as of 2024.10, 10 materials systems)*

```
▼ system specific vocabulary : {  
  ▶ system_catalyst : { 6 props }  
  ▶ system_porous_materials : { 6 props }  
  ▶ system_memristive : { 6 props }  
  ▶ system_organic_thin_film_transistor : { 6 props }  
  ▶ system_photodetector : { 6 props }  
  ▶ system_gas_sensor : { 6 props }  
  ▶ system_alkali_ion_battery : { 6 props }  
  ▶ system_piezoelectric : { 5 props }  
  ▶ system_multilayer : { 6 props }  
  ▶ system_photovoltaic : { 6 props }  
}
```

```
▼ system_gas_sensor : {  
  eng_definition : a device that detects the presence of gases  
  alias : chemoresistive sensor; chemical sensor  
  ▶ description : { 3 props }  
  ▼ configuration : {  
    eng_definition : configurational information of the gas sensor  
    ▶ substrate : { 4 props }  
    ▶ electrode : { 7 props }  
    ▶ channel : { 5 props }  
    ▶ interlayer : { 5 props }  
  }  
  ▶ process : { 2 props }  
  ▼ performance : {  
    eng_definition : performance data of the gas sensor  
    ▶ measurement condition : { 9 props }  
    ▶ responsivity : { 4 props }  
    ▶ recovery time : { 4 props }  
    ▶ response time : { 4 props }  
    ▶ theoretical detection limit : { 4 props }  
    ▶ selectivity : { 3 props }  
    ▶ air stability : { 3 props }  
    ▶ humidity stability : { 4 props }  
    ▶ measurement : { 4 props }  
  }  
}
```

Process Data Construction *(as of 2024.10, 53 unit processes)*

```
▼ common vocabulary for materials process : {  
  ▶ atomic layer deposition : { 13 props }  
  ▶ atomic layer etching : { 13 props }  
  ▶ ball milling : { 16 props }  
  ▶ blade coating : { 7 props }  
  ▶ bulk metal forming : { 10 props }  
  ▶ casting : { 9 props }  
  ▶ centrifugation : { 8 props }  
  ▶ chemical mechanical polishing : { 12 props }  
  ▶ chemical bath deposition : { 7 props }  
  ▶ chemical synthesis : { 7 props }  
  ▶ chemical vapor deposition : { 11 props }  
  ▶ diamond turning : { 5 props }  
  ▶ drying : { 6 props }  
  ▶ dialysis : { 7 props }  
  ▶ e-beam lithography : { 11 props }  
  ▶ electrochemical deposition : { 13 props }  
  ▶ electrospinning : { 11 props }  
  ▶ electrical poling : { 8 props }  
  ▶ encapsulation : { 5 props }  
  ▶ etching : { 4 props }  
  ▶ cleaning : { 4 props }  
  ▶ heat treatment : { 8 props }  
  ▶ hydrothermal reaction : { 9 props }  
  ▶ ion intercalation : { 6 props }  
  ▶ ink jet printing : { 6 props }  
  ▶ metal additive manufacturing : { 22 props }  
  ▶ microwave-assisted synthesis : { 10 props }  
  ▶ mixing : { 7 props }  
  ▶ molecular beam epitaxy : { 11 props }  
  ▶ perovskite postprocess : { 4 props }  
  ▶ polishing : { 12 props }  
  ▶ pressing : { 5 props }  
  ▶ pulsed laser deposition : { 12 props }  
  ▶ quenching : { 4 props }  
  ▶ rinsing : { 5 props }  
  ▶ sintering : { 7 props }  
  ▶ slurry casting : { 8 props }  
  ▶ slot coating : { 4 props }  
  ▶ solvent annealing : { 5 props }
```

```
▼ heat treatment : {  
  eng_definition : thermal process for the purpose of improving the properties of a  
                    material, removing impurities(volatile substances), and/or incur thermal  
                    decomposition  
  
  alias : value  
  ▶ method : { 4 props }  
  ▶ atmosphere : { 4 props }  
  ▶ temperature : { 4 props }  
  ▶ heating rate : { 4 props }  
  ▶ time : { 4 props }  
  ▶ cooling rate : { 4 props }  
}
```

```
▼ microwave-assisted synthesis : {  
  eng_definition : microwave-assisted synthesis conditions that can be controlled during  
                    process  
  ▶ precursor_(n) : { 3 props }  
  ▶ solution_(n) : { 3 props }  
  ▶ heating rate : { 5 props }  
  ▶ temperature : { 4 props }  
  ▶ cooling rate : { 5 props }  
  ▶ atmosphere : { 3 props }  
  ▶ time : { 4 props }  
  ▶ microwave power : { 5 props }  
  ▶ microwave frequency : { 5 props }  
}
```

Analysis Data Construction *(as of 2024.10, 63 analysis methods)*

▼ common vocabulary for analysis method : {

- ▶ AES : { 4 props }
- ▶ AFM : { 4 props }
- ▶ amperometry : { 6 props }
- ▶ atom probe tomography : { 4 props }
- ▶ compressive test : { 8 props }
- ▶ creep test : { 4 props }
- ▶ d33 meter : { 5 props }
- ▶ Dynamic Light Scattering(DLS) : { 5 props }
- ▶ DFT : { 22 props }
- ▶ DMA : { 4 props }
- ▶ differential scanning calorimetry (DSC) : { 4 props }
- ▶ EDS : { 5 props }
- ▶ EIS : { 8 props }
- ▶ electrochemical activity : { 7 props }
- ▶ empirical MD : { 6 props }
- ▶ EPMA : { 4 props }
- ▶ ERD : { 4 props }
- ▶ EQE analysis : { 5 props }
- ▶ EXAFS : { 4 props }
- ▶ fatigue test : { 13 props }
- ▶ film adhesion test : { 5 props }
- ▶ focussed ion beam : { 5 props }
- ▶ fsLA-ICP-MS : { 5 props }
- ▶ gas adsorption/desorption isotherm : { 8 props }
- ▶ gas chromatography : { 10 props }
- ▶ gel electrophoresis : { 6 props }
- ▶ liquid chromatography : { 4 props }
- ▶ hardness test : { 5 props }
- ▶ ICP-MS : { 5 props }
- ▶ ICP-OES : { 5 props }
- ▶ impact test : { 4 props }
- ▶ impedance analyzer : { 4 props }
- ▶ infrared spectroscopy : { 6 props }
- ▶ IV measurement : { 6 props }
- ▶ JV measurement : { 16 props }
- ▶ magnetic hysteresis loop : { 7 props }
- ▶ memristive activity : { 8 props }
- ▶ nano-indentation : { 4 props }
- ▶ NEXAFS : { 4 props }
- ▶ nuclear magnetic resonance : { 5 props }

▼ DFT : {

- eng_definition : quantum mechanical calculations using density functional theory
- ▶ code : { 4 props }
- ▶ calculation mode : { 4 props }
- ▶ basis : { 3 props }
- ▶ charge : { 5 props }
- ▶ energy cutoff : { 5 props }
- ▶ optimizer : { 4 props }
- ▶ exchange-correlation functional : { 4 props }
- ▶ solvent model : { 3 props }
- ▶ k-point : { 3 props }
- ▶ convergence criteria : { 4 props }
- ▶ symmetry : { 4 props }
- ▶ magnetic structure : { 4 props }
- ▶ potential : { 4 props }
- ▶ LDA+U : { 6 props }
- ▶ partial occupation : { 4 props }
- ▶ total energy : { 5 props }
- ▶ density of state : { 6 props }
- ▶ phonon calculations : { 7 props }
- ▶ nudged elastic band : { 7 props }
- ▶ ab-initio MD : { 7 props }
- ▶ spin orbit coupling : { 4 props }

▼ JV measurement : {

- eng_definition : IV measurement of solar cell as it is illuminated
- alias : value
- ▶ equipment : { 3 props }
- ▶ stabilized test : { 5 props }
- ▶ number of cells averaged : { 5 props }
- ▶ certified : { 4 props }
- ▶ certification institute : { 4 props }
- ▶ storage before measurement : { 5 props }
- ▶ atmosphere : { 4 props }
- ▶ relative humidity : { 5 props }
- ▶ temperature : { 5 props }
- ▶ light source : { 5 props }
- ▶ light : { 8 props }
- ▶ scan : { 8 props }
- ▶ preconditioning : { 6 props }
- ▶ raw data : { 4 props }

Allowed Units based on 7 basic SI units

rad = 7 SI base units

Time		Length		Mass	
year	year	m	meter	kg	kilogram
d	day	cm	centi meter	g	gram
h	hour	mm	milli meter	mg	milli gram
min	minute	{Wmu m}	micro meter	m_{e}	electron mass
s	second	nm	nano meter	Da	dalton
ms	milli second	WANGSTROM	Angstrom		
{Wmu s}	micro second				
ns	nano second				
fs	femto second				

Electricity		Thermodynamic Temperature / Energy / Power		Amount	
A	Ampere	eV	electron volt	mol	mole
mA	milli Ampere	keV	kilo electron volt		
{Wmu A}	micro Ampere	meV	milli electron volt		
nA	nano Ampere	J	Joule		
pA	pico Ampere	kJ	kilo Joule		
V	volt	pJ	pico Joule		
kV	kilo volt	W	watt		
mV	millivolt	K	kelvin degree		
{Wmu V}	micro volt	kW	kilo watt		
C	coulomb	MeV	mega electron volt		
{Wmu C}	micro Coulomb	mW	milli watt		
pC	pico Coulomb	{Wmu W}	micro watt		
e	charge of an electron				
S	conductance (Siemens)				
Ohm	resistance				
F	Faraday				

Luminous Intensity		Force / Pressure		Others	
cd	candela	N	Newton	at. %	atomic per cent
		{Wmu N}	micro Newton	wt. %	weight per cent
		kN	kilo Newton	ppm	part per million
		kgf	kilo gram force	%	per cent
		gf	gram force	HR	Hardness scale
		Pa	pascal	HB	
		MPa	mega pascal	HV	
		GPa	giga pascal	HK	
		Torr	Torr	Jones	specific detectivity
		bar	Bar	Hz	Hertz
		atm	atmosphere	kHz	kilo Hertz
				revolution	revolution, rotation
				degree	geometric angle
				L	liter
				mL	milli liter
				cycle	number of repeat
				T	tesla
				{Wmu L}	micro liter
				GHz	giga Hertz

소재 연구데이터 표준어휘 사전 작성
가이드

Guide for Building the Standard Vocabulary
Dictionary of Materials Research Data

2024.11.1.

소재 연구데이터 표준화 전문위원회

MatResData-Standard-Committee in github.com